

MCQ Mock Test for 41107 Marine and Ocean Engineering

[Practice examination set generated from the uploaded mock test, homework notes, transcribed examples, and the updated formula sheet]

The scoring algorithm is based on **one best answer**.

This means:

- There is always only one correct answer.
- Students are only able to select one answer per question.
- Every correct answer corresponds to 1 point.
- Every incorrect answer corresponds to 0 points.

If not specifically stated, you may assume the following:

- Seawater density $\rho = 1025 \text{ kg/m}^3$ and freshwater density $\rho = 1000 \text{ kg/m}^3$.
- Kinematic viscosity $\nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$.
- Deep-water conditions and long-crested waves where appropriate.
- Wave and response processes are Gaussian and peak values follow Rayleigh statistics where appropriate.

Mock exam

Part I — Flow around cylinders, VIV, and oscillatory-flow thinking

Question 1

FLOW AROUND AND FORCES ON CYLINDERS. Steady current around a free smooth circular cylinder. Consider a cylinder with $D = 1.0 \text{ m}$ in a current $U_c = 0.25 \text{ m/s}$. Take $\nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$. Which flow regime is most appropriate?

- A. Sub-critical
- B. Critical / trans-critical onset
- C. Super-critical
- D. No vortex shedding because $Re < 1$

Correct answer: A. $Re = U_c D / \nu = 0.25 \cdot 1.0 / 10^{-6} = 2.5 \times 10^5$. This lies in the sub-critical range used in the course, so the correct answer is **A**.

Question 2

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. Consider the following three cases. Assume $C_D = 1.2$ in the sub-critical regime and $C_D = 0.9$ in the trans-critical regime. Which case gives the largest mean drag force per unit length?

- A. $D = 0.3 \text{ m}$, $U_c = 0.4 \text{ m/s}$
- B. $D = 4.0 \text{ m}$, $U_c = 0.25 \text{ m/s}$
- C. $D = 0.8 \text{ m}$, $U_c = 0.6 \text{ m/s}$

Correct answer: C. Use $F_D = \frac{1}{2}\rho C_D D U_c^2$. Case A: $Re = 1.2 \times 10^5$ so $C_D = 1.2$, giving a factor $1.2 \cdot 0.3 \cdot 0.4^2 = 0.0576$. Case B: $Re = 1.0 \times 10^6$ so $C_D = 0.9$, factor $0.9 \cdot 4.0 \cdot 0.25^2 = 0.225$. Case C: $Re = 4.8 \times 10^5$ so $C_D = 0.9$, factor $0.9 \cdot 0.8 \cdot 0.6^2 = 0.2592$. Case C is largest, so **C**.

Question 3

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder. What is the pressure in excess of hydrostatic pressure at the stagnation point if $U_c = 1.6$ m/s and $\rho = 1025$ kg/m³?

- A. 0 Pa
- B. 820 Pa
- C. 1310 Pa
- D. 2620 Pa

Correct answer: C. At the stagnation point, $p_s - p_0 = \frac{1}{2}\rho U_c^2$. So $p_s - p_0 = 0.5 \cdot 1025 \cdot 1.6^2 = 1312$ Pa $\approx$ 1310 Pa. Therefore **C**.

Question 4

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder is described by $U(t) = U_m \sin(\omega t)$ with $U_m = 0.9$ m/s, $T = 6.0$ s and $D = 0.60$ m. What is the Keulegan–Carpenter number?

- A. $KC = 6.0$
- B. $KC = 9.0$
- C. $KC = 12.0$
- D. $KC = 15.0$

Correct answer: B. $KC = U_m T / D = (0.9 \cdot 6.0) / 0.60 = 9.0$. Hence **B**.

Question 5

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted circular cylinder. The oscillatory flow is $U(t) = 1.8 \sin(\omega t)$ m/s, the diameter is $D = 0.40$ m, and the angular natural frequency is $\omega_n = 9.42$ rad/s. What is the reduced velocity V_r ?

- A. $V_r = 2.0$
- B. $V_r = 3.0$
- C. $V_r = 4.5$
- D. $V_r = 6.0$

Correct answer: B. $f_n = \omega_n / (2\pi) = 9.42 / (2\pi) \approx 1.50$ Hz. Then $V_r = U_m / (D f_n) = 1.8 / (0.40 \cdot 1.50) = 3.0$. So **B**.

Question 6

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. For $U_m = 0.8 \text{ m/s}$, $D = 0.50 \text{ m}$ and $T = 4.0 \text{ s}$, assume $C_D = 1.0$ and $C_M = 1.8$. What is the ratio $F_{I,\max}/F_{D,\max}$ approximately?

- A. 0.28
- B. 1.39
- C. 2.78
- D. 5.56

Correct answer: C. First $KC = U_m T / D = 0.8 \cdot 4.0 / 0.50 = 6.4$. Then $F_{I,\max}/F_{D,\max} = (\pi^2 / KC)(C_M / C_D) = \pi^2 \cdot 1.8 / 6.4 \approx 2.78$. So **C**.

Question 7

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. The normalized fundamental lift frequency is $N_L = 2$. Which name is given to this flow regime?

- A. Single-pair regime
- B. Double-pair regime
- C. Three-pair regime
- D. No-separation regime

Correct answer: A. In the lecture convention, $N_L \approx 2$ corresponds to the **single-pair regime**. So the correct answer is **A**.

Question 8

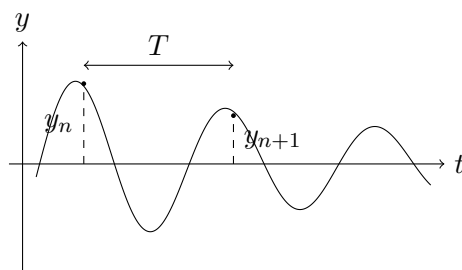
FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. If the vortex shedding frequency is $f_v = 0.24 \text{ Hz}$, what is the frequency of the oscillating drag force?

- A. 0.12 Hz
- B. 0.24 Hz
- C. 0.48 Hz
- D. 0.96 Hz

Correct answer: C. For a cylinder wake, the oscillating drag force occurs at $2f_v$. Thus $f_{D,\text{osc}} = 2 \cdot 0.24 = 0.48 \text{ Hz}$. Hence **C**.

Question 9

VORTEX INDUCED VIBRATIONS. The amplitude response in a free-decay test is shown in the figure. If $y_n = 1.5$, $y_{n+1} = 0.9$ and $T = 0.8 \text{ s}$, what are the damping ratio ζ and the undamped natural angular frequency ω_n approximately?



- A. $\zeta = 0.081$, $\omega_n = 7.88 \text{ rad/s}$
- B. $\zeta = 0.081$, $\omega_n = 6.28 \text{ rad/s}$
- C. $\zeta = 0.163$, $\omega_n = 7.88 \text{ rad/s}$
- D. $\zeta = 0.163$, $\omega_n = 6.28 \text{ rad/s}$

Correct answer: A. Using the logarithmic decrement approximation, $\zeta \approx \frac{1}{2\pi} \ln(y_n/y_{n+1}) = \frac{1}{2\pi} \ln(1.5/0.9) = 0.0813$. The damped frequency is $\omega_d = 2\pi/T = 2\pi/0.8 = 7.85 \text{ rad/s}$. Then $\omega_n = \omega_d/\sqrt{1 - \zeta^2} \approx 7.88 \text{ rad/s}$. So **A**.

Question 10

VORTEX INDUCED VIBRATIONS. Calculate the undamped natural frequency of a flexibly mounted circular cylinder in water. Take $D = 0.08 \text{ m}$, structural mass per unit length $m = 4.0 \text{ kg/m}$, spring constant per unit length $k = 900 \text{ N/m}$, and assume $A/D < 0.8$ so that $c_m = 1$ for the added mass estimate. What is f_n approximately?

- A. 0.79 Hz
- B. 1.58 Hz
- C. 2.25 Hz
- D. 3.16 Hz

Correct answer: B. Use $m' = \rho c_m A = \rho A$ with $A = \pi(D/2)^2 = \pi(0.04)^2 = 0.00503 \text{ m}^2$. So $m' \approx 1025 \cdot 0.00503 = 5.15 \text{ kg/m}$. Then $f_n = \frac{1}{2\pi} \sqrt{k/(m + m')} = \frac{1}{2\pi} \sqrt{900/(4.0 + 5.15)} \approx 1.58 \text{ Hz}$. Therefore **B**.

Question 11

DIFFRACTION / LARGE BODIES. For wave diffraction around a cylinder, when is diffraction expected to be important?

- A. When $D/L \gtrsim 0.2$
- B. When $D/L \lesssim 0.02$
- C. Only when $KC > 20$
- D. Only when $Re < 1000$

Correct answer: A. In the lecture notes, diffraction effects become important for relatively large bodies, typically when $D/L \gtrsim O(0.2)$. Hence **A**.

Question 12

FLOW AROUND AND FORCES ON CYLINDERS. For a smooth circular cylinder in the sub-critical regime, why is C_D almost constant over a wide range of Re ?

- A. Because separation occurs early and at nearly the same angular position, so the wake width stays similar.
- B. Because friction drag dominates completely and pressure drag is negligible.
- C. Because the boundary layer is turbulent from the stagnation point.
- D. Because vortex shedding stops in the sub-critical regime.

Correct answer: A. In the sub-critical regime, separation is early and occurs at nearly the same angular position over a wide range of Re , so the wake width and pressure drag remain similar. That is why C_D stays almost constant. Therefore **A**.

Part II — Ship motion, hydrostatics, resistance, spectra, and strength**Question 13**

SHIP MOTION DYNAMICS. Transfer functions of a ship are available. At a given frequency, the real and imaginary parts of the pitch RAO are $(\text{Re}, \text{Im}) = (-4.0 \times 10^{-2}, 3.0 \times 10^{-2})$ rad/m. What is the phase angle ε of the complex RAO, expressed in degrees in the interval $[0, 360)$?

- A. 36.9°
- B. 143.1°
- C. 216.9°
- D. 323.1°

Correct answer: B. The point lies in quadrant II because $\text{Re} < 0$ and $\text{Im} > 0$. The reference angle is $\alpha = \tan^{-1}(|0.03/0.04|) = 36.9^\circ$. So $\varepsilon = 180^\circ - \alpha = 143.1^\circ$. Hence **B**.

Question 14

FORWARD SPEED / ENCOUNTERED WAVES. A ship travels at 18 knots in head seas. The true wave frequency is $f = 0.10$ Hz. What is the encountered period T_e approximately?

- A. 6.3 s
- B. 7.2 s
- C. 8.5 s
- D. 10.0 s

Correct answer: A. Convert speed: $U = 18 \cdot 0.51444 = 9.26$ m/s. Then $\omega = 2\pi f = 0.628$ rad/s. For head seas, $\omega_e = \omega + \omega^2 U/g = 0.628 + 0.628^2 \cdot 9.26/9.81 = 1.001$ rad/s. So $T_e = 2\pi/\omega_e = 6.28$ s ≈ 6.3 s. Therefore **A**.

Question 15

SHIP MOTION DYNAMICS. A vertical circular cylinder floats with its axis vertical and undergoes free oscillations in pure heave. Damping is neglected. The draught measured from the bottom of the cylinder is $h = 0.80$ m. What is the natural angular frequency approximately?

- A. 2.5 rad/s
- B. 3.5 rad/s
- C. 4.5 rad/s
- D. 5.5 rad/s

Correct answer: B. For the simplified vertical-cylinder heave model, $\omega_n = \sqrt{g/h}$. Thus $\omega_n = \sqrt{9.81/0.80} = 3.50$ rad/s. So **B**.

Question 16

SPECTRAL ANALYSIS. At a frequency ω_a , the wave energy spectral density is $S(\omega_a) = 1.8 \text{ m}^2\text{s/rad}$ and the corresponding pitch response spectral density is $R(\omega_a) = 0.008 \text{ rad}^2\text{s/rad}$. What is the magnitude of the pitch transfer function $|\Phi(\omega_a)|$ approximately?

- A. 0.0667 rad/m (3.8 deg/m)
- B. 0.0383 rad/m (2.2 deg/m)
- C. 0.0067 rad/m (0.38 deg/m)
- D. 0.1067 rad/m (6.1 deg/m)

Correct answer: A. Use $R(\omega) = |\Phi(\omega)|^2 S(\omega)$. So $|\Phi| = \sqrt{R/S} = \sqrt{0.008/1.8} = 0.0667$ rad/m. In degrees per metre this is $0.0667 \cdot 180/\pi = 3.82$ deg/m. Hence **A**.

Question 17

HYDROSTATIC COEFFICIENTS. A ship in saltwater has $L = 120$ m, $B = 20$ m, $T = 8$ m and displacement mass $\Delta = 15000$ t. Taking $\rho = 1.025 \text{ t/m}^3$, what is the block coefficient C_B approximately?

- A. 0.56
- B. 0.68
- C. 0.76
- D. 0.91

Correct answer: C. The displaced volume is $\nabla = \Delta/\rho = 15000/1.025 = 14634 \text{ m}^3$. Then $C_B = \nabla/(LBT) = 14634/(120 \cdot 20 \cdot 8) = 0.762$. Therefore **C**.

Question 18

SHIP HYDROSTATICS. A rectangular, box-shaped barge has beam $B = 8.0$ m and floats at uniform draught T . What must the draught be for the transverse metacentre to lie in the waterplane?

- A. $T = 2.5$ m
- B. $T = 3.3$ m

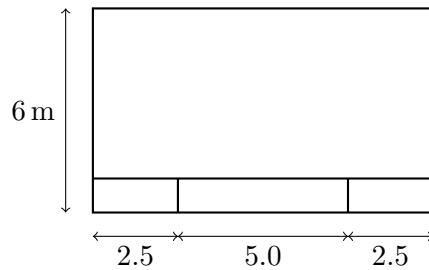
C. $T = 4.0$ m

D. $T = 4.6$ m

Correct answer: B. For a rectangular barge with M_T in the waterplane, $T = B/\sqrt{6}$. So $T = 8/\sqrt{6} = 3.27$ m ≈ 3.3 m. Hence **B**.

Question 19

STRENGTH OF SHIPS. The thin-walled barge cross section shown below has uniform plate thickness $t = 10$ mm. What is the vertical distance z_n from the keel to the neutral axis?



A. $z_n = 2.32$ m

B. $z_n = 2.43$ m

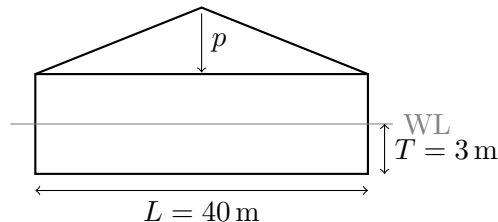
C. $z_n = 2.57$ m

D. $z_n = 3.00$ m

Correct answer: B. With uniform thickness, use $z_n = \frac{\sum l_i z_i}{\sum l_i}$. The plate lengths are 10, 10, 10, 6, 6, 1, 1 m, with centroid heights 0, 1, 6, 3, 3, 0.5, 0.5 m. Thus $\sum l_i z_i = 0 + 10 + 60 + 18 + 18 + 0.5 + 0.5 = 107$ and $\sum l_i = 44$. So $z_n = 107/44 = 2.43$ m. Therefore **B**.

Question 20

STRENGTH OF SHIPS. The barge below is loaded with gravel. The gravel load is triangular along the whole length with maximum intensity p at midship. The barge's own weight is uniformly distributed with intensity $0.2p$. The barge length is 40 m, breadth is 6 m, and it floats at draught $T = 3$ m in freshwater. Approximately what value has p ?



A. $p = 18$ tons/m

B. $p = 26$ tons/m

C. $p = 32$ tons/m

D. $p = 40$ tons/m

Correct answer: B. In freshwater, displaced weight is approximately $LBT = 40 \cdot 6 \cdot 3 = 720$ tons. The triangular gravel load has area $\frac{1}{2} \cdot 40 \cdot p = 20p$. The barge own weight contributes $40 \cdot 0.2p = 8p$. Total load $= 28p = 720$, so $p = 25.7$ tons/m ≈ 26 tons/m. Hence **B**.

Question 21

SHIP RESISTANCE. A ship of length 180 m is represented by a model 4.0 m long for resistance estimation. If performance is compared at corresponding speeds, what is the ratio R_s/R_m approximately?

- A. 9.1×10^3
- B. 9.1×10^4
- C. 9.1×10^5
- D. 4.5×10^4

Correct answer: B. At corresponding speeds, resistance scales approximately with α^3 , where $\alpha = L_s/L_m = 180/4 = 45$. Thus $R_s/R_m \approx 45^3 = 91125 \approx 9.1 \times 10^4$. So **B**.

Question 22

SHIP RESISTANCE / MODEL TESTING. In towing-tank resistance estimation for ships with free-surface effects, what similarity must be preserved in practice?

- A. Identical Froude numbers of model and ship
- B. Identical Reynolds numbers of model and ship
- C. Identical Froude and Reynolds numbers simultaneously
- D. Identical Strouhal numbers only

Correct answer: A. For ship resistance with free-surface effects, model testing preserves **Froude similarity**. Exact Reynolds similarity cannot be achieved simultaneously for ship and model. Therefore **A**.

Question 23

WAVE STATISTICS / SPECTRAL MOMENTS. A wave system has $H_s = 4.0$ m and first spectral moment $m_1 = 0.80 \text{ m}^2 \text{ rad/s}$. What is a representative mean period T_1 approximately?

- A. 6.3 s
- B. 7.9 s
- C. 9.4 s
- D. 12.6 s

Correct answer: B. First find $m_0 = (H_s/4)^2 = (4.0/4)^2 = 1.0 \text{ m}^2$. Then $T_1 = 2\pi m_0/m_1 = 2\pi \cdot 1.0/0.80 = 7.85 \text{ s}$. So the best answer is **B**.

Question 24

WAVE STATISTICS. For an irregular wave record, the variance of the surface elevation is $m_0 = 1.44 \text{ m}^2$. What is the probability that a wave crest exceeds the level $h_1 = 1.2 \text{ m}$?

- A. 13.5%
- B. 36.8%
- C. 60.7%
- D. 77.9%

Correct answer: C. Use the Rayleigh crest exceedance formula $P(Z > h_1) = \exp[-h_1^2/(2m_0)]$. Thus $P = \exp[-1.2^2/(2 \cdot 1.44)] = \exp(-0.5) = 0.607$, i.e. 60.7%. Therefore **C**.